KENYA MEDICAL TRAINING COLLEGE
FACULTY OF INFORMATION TECHNOLOGY AND MEDICAL EDUCATION
DEPARTMENT OF HEALTH RECORDS AND INFORMATION TECHNOLOGY

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

PAPER: HEALTH STATISTICS

DATE: 27th JUNE, 2023 TIME: 3 HOURS (9:00 – 12:00 NOON)

INSTRUCTIONS

1. *This paper consists of:*
   - *Section 1* (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. ALL Multiple choice questions MUST be answered in **CAPITAL LETTERS**
4. Write your **EXAMINATION NUMBER** on all the answer sheets provided and on question paper.

...your Examination Numbe

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. ABC company wants to test the effectiveness of a new advertisement on different age groups. They divide the target population into five age categories and randomly select participants from each group. Which sampling method is used?
   A. Simple random sampling
   B. Systematic sampling
   C. Stratified random sampling
   D. Cluster sampling
2. Which type of statistics is used to draw conclusions about a population based on a sample?
   A. Descriptive statistics
   B. Inferential statistics
   C. Exploratory statistics
   D. Correlational statistics
3. A hospital records the time it takes for their customer service representatives to respond to the patient's inquiries. Which of the following best describes the type of data being collected?
   A. Discrete data
   B. Continuous data
   C. Nominal data
   D. Ordinal data
4. A researcher asks participant to rate their satisfaction with a product on a scale of 1 to 5 with 1 being very unsatisfied and 5 being very satisfied, which type of measurement scale is being used?
   A. Nominal scale
   B. Ordinal scale
   C. Interval scale
   D. Ratio scale
5. A researcher is conducting a study on the effects of a new drug for Covid-19, which of the following sources would provide primary data
   A. Medical textbooks
   B. Clinical results published in a journal
   C. Patient interview conducted by research
   D. Expert opinions from health care professionals
6. A research wants to gather data on customer preferences for a new product. Which da collection method would be most appropriate to obtain in-depth insights and opinions?
   A. Questionnaire method
   B. Observation method
   C. Focus group discussion
   D. Interview method
7. The discrepancy between the sample characteristics and the population characterist referred to as?
   A. Sampling bias
   B. Sampling discrepancy

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C. Sampling error
D. Selection bias

8. A researcher wants to comprehend sales performance of different product categories over time, which data presentation method would be most appropriate?
   A. Bar chart
   B. Line-graph
   C. Pie chart
   D. Scatter plot
9. Which term best describes the total number of patients remaining in the hospital or ward each day added together over the reference period?
   A. Occupied bed days
   B. Turnover per bed
   C. Available bed days
   D. Daily occupancy
10. A company wants to compare the sales across multiple regions. Which data presentation method would be most suitable for this comparison?
   A. Tabular form
   B. Bar chart
   C. Histogram
   D. Scatter plot
11. Which data presentation method displays the individual items as collected in order of their magnitude that is from the lowest to the highest and vice versa?
   A. Bar chart
   B. Histogram
   C. Array
   D. Lorenz curve
12. The range between the first quartile (Q1) and third quartile (Q3) is called?
   A. Interquartile range
   B. Quartile ray
   C. Semi-interquartile range
   D. Quartile deviation
13. Which quartile represents the median of a given dataset
   A. First quartile (Q1)
   B. Second quartile (Q2)
   C. Third Quartile (Q3)
   D. Fourth Quartile (Q4)
14. Which term best describes the average number of patients expected to have been treated on a bed in a specified period?
   A. Turn over per bed
   B. Occupied bed days
   C. Turnover interval
   D. Average length of stay
15. Deciles divide a database into how many parts
   A. Two equal parts

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B A hundred equal parts
C Four equal parts
D Ten equal parts

16. Which of the following measures is a relative measure of dispersion?
A Range
B Variance
C Standard deviation
D Coefficient of variation

17. Which measure of central tendency can be used to compare the average rate of growth?
A Harmonic mean
B Arithmetic mean
C Weighted mean
D Geometric mean

18. In a moderately skewed distribution, the median is 40 and the mean is 42, using these values what is the approximate value of the mode
A 30
B 46
C 72
D 18

19. The mean of ten numbers is 58, if one of the numbers is 40, what is the mean of the other nine?
A 18
B 60
C 162
D 540

20. Which measure of a central tendency is calculated by dividing the total number of all observation by the number of items?
A Arithmetic mean
B Geometric mean
C Median
D Harmonic mean

21. What is the median of this sample? 5,5,11,9,8,5,8
A 5
B 6
C 8
D 9

22. What is the mean of the following items, 20,50,80,30,20?
A 20
B 40
C 30
D 50

23. What is the mode of this dataset,- 2,6,5,8,6,10
A 2
B 4

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C. 6
D. 8

24. In a grouped frequency distribution, how can the modal group be determined?
    A. Identifying the class with the highest frequency
    B. Calculating the mean of all the classes
    C. Taking the median of the class boundaries
    D. Summing all the class frequency and diverting
25. Which method of data presentation shows the mode graphically?
    A. Ogive
    B. Bar chart
    C. Histogram
    D. Line curve
26. Which method of data presentation shows the median graphically?
    A. Ogive
    B. Barchart
    C. Histogram
    D. Line curve
27. What represents the mode in a bimodal dataset?
    A. The average of the two highest values in the dataset
    B. The difference between the highest and lowest values in the dataset
    C. The 2 values that appear most frequently in the dataset
    D. The middle value of the dataset
28. Which measure takes the relevant root of the product of all non-zero and positive values?
    A. Weighted mean
    B. Geometric mean
    C. Harmonic mean
    D. Arithmetic mean
29. Which is the appropriate average for calculating the average speed of a journey
    A. Weighted mean
    B. Geometric mean
    C. Harmonic mean
    D. Arithmetic mean
30. In a grouped frequency distribution, how is the harmonic mean determined?
    A. Calculate the reciprocal of the weighted mean
    B. Taking the median of the class boundaries
    C. Identifying the class with the highest frequency
    D. Summing all the class frequencies and dividing by the number of classes
31. The sum of the deviation of a variable from its mean is always equal to?
    A. 0
    B. 1
    C. 2
    D. 5
32. If the arithmetic mean of 20 values is 10, then sum of these 20 values is?
    A. 10

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B. 20
C. 200
D. 20+10

33. When the value of the mean is greater than the mode the skewness is said to be?
A. Negatively skewed
B. Positively skewed
C. Symmetrical
D. Mean skewed

34. A list of all items within a population from which we want to do sampling is known as
A. Sampling frame
B. Sample size
C. Sample
D. Sampling technique,

35. What is the standard deviation of the following series; 5,8,7,11,14
A. 3.24
B. 3.16
C. 3.53
D. 3.65

36. The occupied bed days of XYZ hospital for the month of March were 6,200 patients, what was the average daily occupancy for the period?
A. 100
B. 200
C. 16.98
D. 620

37. What is the range of the following scores, 20, 145, 27, 240?
A. 220
B. 236
C. 244
D. 27

38. The type of cumulative frequency distribution in which the class intervals are added in bottom to top the order is classified as?
A. More than type distribution
B. Variation distribution
C. Marginal distribution
D. Less than type distribution

39. The measurement of scale which allows the ranking of numbers rather than arithmetic operations on data is classified as?
A. Valid scale
B. Continuous scale
C. Discrete scale
D. Ordinal scale

40. How is a relative frequency distribution obtained?
A. Dividing frequency of each class interval by the total frequency
B. Dividing the frequency of each class interval by the number of class intervals

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C. Dividing the range of data by the total frequency
D. Dividing the frequency of each class interval by the mean, of the dataset

SECTION 2: SHORT ANSWER QUESTIONS (40MARKS

1. Use the following data from a discrete series to compute median (5 marks)

Height (in cm)	160	150	152	161	156	154	155
No. of students	12	8	4	4	3	3	7

2. The mean of the following data is 2.96, compute the mode (5 marks)

x	f
1	3
2	5
3	9
4	a
5	2

3. In X hospital, there were 250 allocated beds OBD of 5,000 patients, 500 discharges in a period of 15 days, use the data to compute and interpret
   a) Percentage occupancy (3 marks)
   b) Average length of stay (2 marks)
4. Define mode and state the four (4) types of modal distributions (5 marks)
5. Enumerate the 5 characteristics of statistics (5 marks)
6. Highlight any five (5) methods of data presentation (5 marks)
7. Outline procedures of carrying out stratified sampling technique (5 marks)
8. State the 5 differences between an interview schedule and a questionnaire (5 marks)

SECTION 3: LONG ANSWER QUESTIONS (20MARKS

9. Use the table below to compute the given measures

Ages in years	no. of Persons
0-10	20
10-20	25
20-30	32
30-40	40
40-50	42
50-60	35
60-70	10
70-80	8

   a) Mean using the arbitrary points method using class (40-50) (3 marks)
   b) Median (3 marks)
   c) Standard deviation (3 marks)
   d) Quartile deviation (3 marks)
   e) Construct a graphical presentation showing mode (8 marks)

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